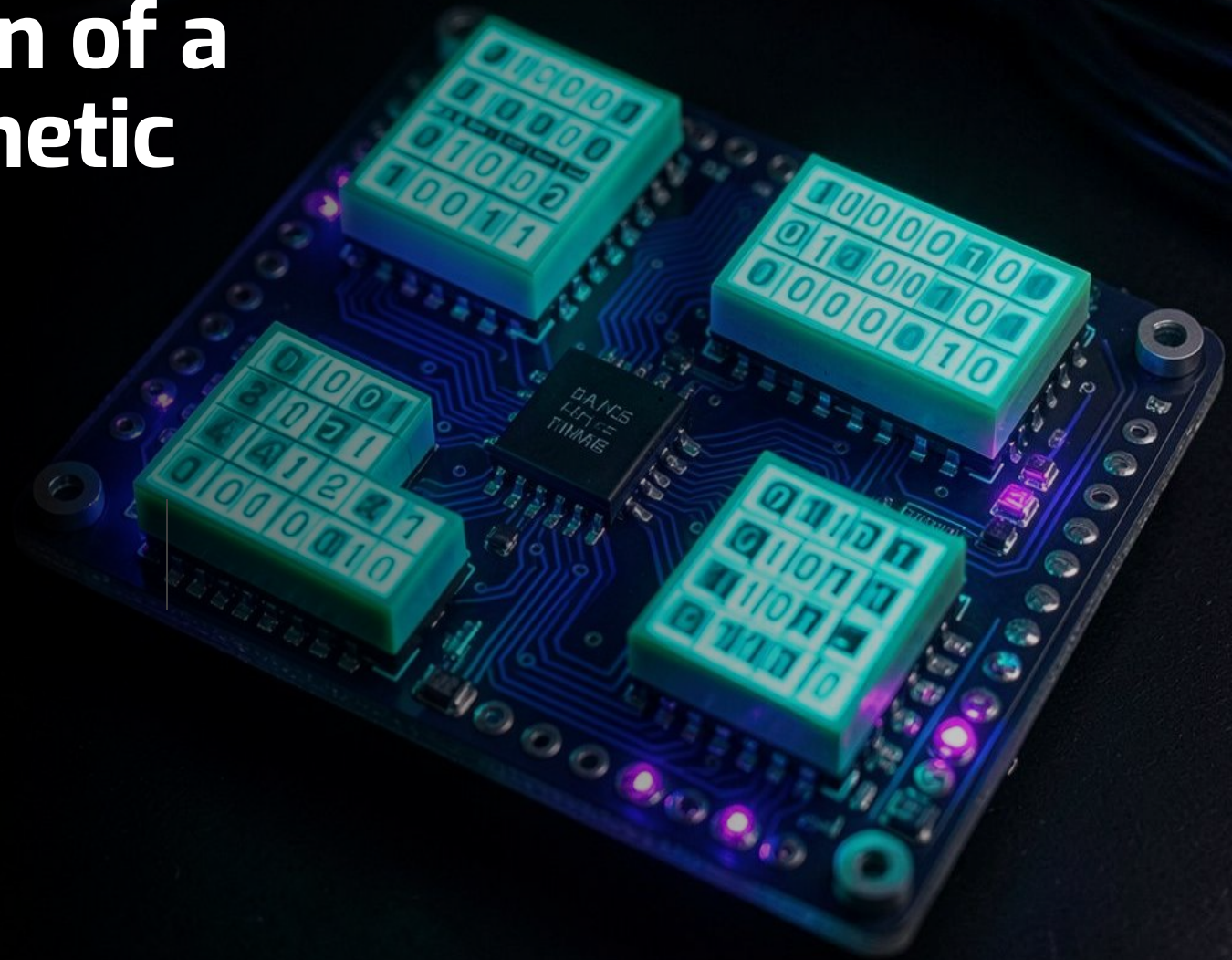
DDCA · MODULE 5 · THE RISC-V PROCESSOR

Design and Stimulation of a Four-Bit Binary Arithmetic Calculator

Using Combinational Logic

TEAM

Batch 1



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01 · WHY THIS BUILDING BLOCK

Binary arithmetic is the smallest useful ALU building block

A four-bit calculator makes processor arithmetic concrete: inputs enter, logic combines them, and a result appears.

01
10

Accept two inputs

Take two four-bit binary values and produce an arithmetic result.



Design with logic

Build the circuit with logic gates and implement it in Logisim Evolution.



Connect to the ALU

See how a small arithmetic circuit becomes one part of a processor datapath.



Place it in RISC-V

RISC-V provides an open instruction set architecture for understanding this context.

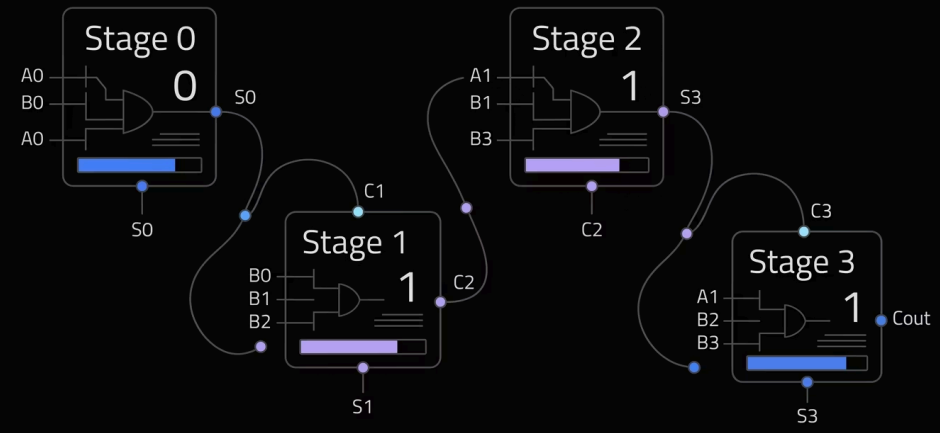
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02 · CIRCUIT DESIGN

Four full adders form a ripple-carry arithmetic circuit

Unsigned four-bit inputs range from **0000** to **1111**. Each stage depends only on the current inputs and its incoming carry.

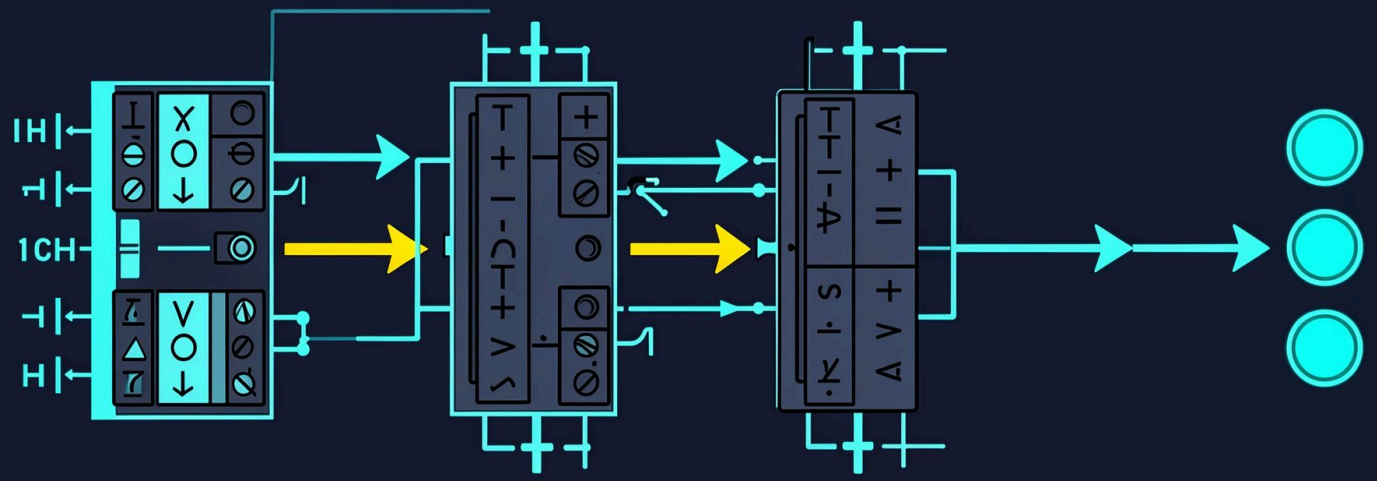
Boolean equations: Sum = A XOR B XOR Cin; Carry = (A AND B) OR (Cin AND (A XOR B)).



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Logisim makes the data path visible from switches to LEDs

The simulator exposes every signal: input switches feed the full-adder stages, carry propagates from the least-significant bit to the most-significant bit, and LEDs display the result.



ILLUSTRATIVE CIRCUIT DIAGRAM · NOT A SUPPLIED SCREENSHOT

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04 · VERIFICATION PLAN

Expected test cases make correctness easy to verify

Use the full-adder truth table to check one stage, then use expected four-bit additions to compare against Logisim outputs. No simulation result is claimed here without supplied evidence.

A	B	CIN	SUM	COUT
0	0	0	0	0
0	1	0	1	0
1	0	1	0	1
1	1	1	1	1

EXPECTED CASE 01

0011 + 0010 = 0101

Actual Logisim output: _____

EXPECTED CASE 02

0111 + 0001 = 1000

Actual Logisim output: _____

EXPECTED CASE 03

1111 + 0001 = 1 0000

Actual Logisim output: _____

The final case needs five result bits, so Cout = 1.

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The project demonstrates one small part of a processor datapath

This project builds four-bit binary addition with combinational logic and uses Logisim for verification when evidence is available.

It is an arithmetic building block, not a complete RISC-V datapath.

In the fetch–decode–execute cycle, instructions such as ADD and SUB use the ALU.

FETCH

Program Counter · instruction memory

DECODE

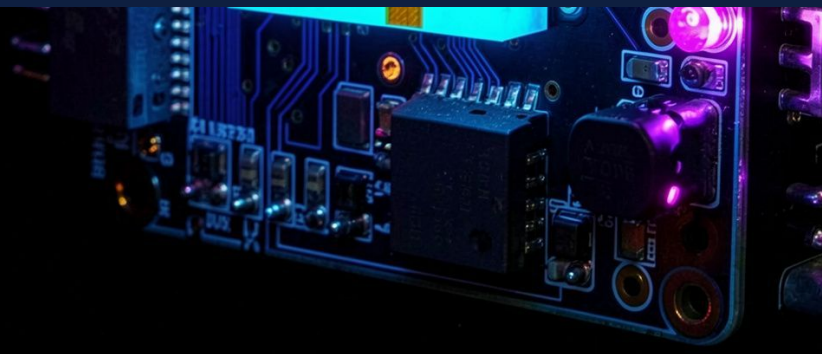
Register file · control unit

EXECUTE

ALU

WRITE BACK

Result · data memory



FUTURE IMPROVEMENTS

Subtraction · more operations · operation selector · seven-segment display · simplified RISC-V datapath integration

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THANK YOU

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